

Guide to Chess PGN Notation and Structure



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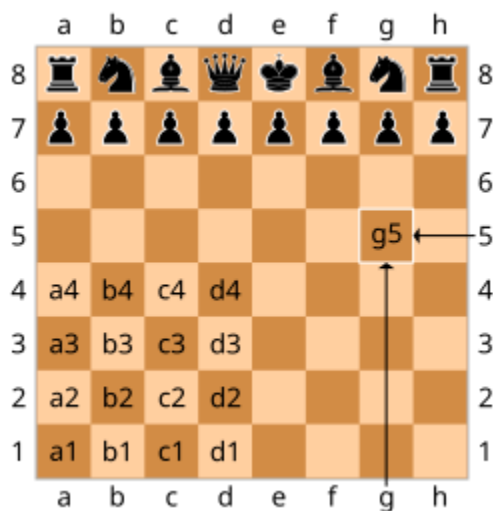
As a chess enthusiast and Python developer, I'll explain Portable Game Notation (PGN) in a clear, step-by-step way. PGN is the standard file format for recording chess games, making it easy to share, analyze, and replay them on computers or apps. It's human-readable (like plain text) but structured enough for software (like chess engines or databases) to parse. Think of it as a "digital scoresheet" for chess.

I'll break this down into sections: what PGN is, its overall structure, detailed explanations of each part, common features like comments and variations, and examples. I'll use simple language, analogies, and real examples to make it descriptive and easy to follow.

1. Chess Board Setup and Nomenclature

Each square of the board is identified by a unique coordinate pair—a letter and a number—from White's point of view. The vertical columns of squares, called *files*, are labeled *a* through *h* from White's left (the queenside) to right (the kingside). The horizontal rows of squares, called *ranks*, are numbered *1* to *8* starting from White's side of the board. Thus each square has a unique identification of file letter followed by rank number. For example, the initial square of White's king is designated as "e1".

Each piece type (other than pawns) is identified by an uppercase letter. English-speaking players use the letters *K* for king, *Q* for queen, *R* for rook, *B* for bishop and *N* for knight. Different initial letters are used in other languages.



2. What is PGN and Why Use It?

- **Definition:** PGN stands for **Portable Game Notation**. It's a text-based format to store chess games, including moves, player names, results, and optional details like dates or comments.

- **Analogy:** Imagine a chess game as a story. PGN is like a book: the "headers" are the title page (who, when, where), and the "movetext" is the story itself (the sequence of moves).
- **Key Benefits:**
 - **Portability:** Works across platforms (e.g., Chess.com, Lichess, or your own Python program).
 - **Compact:** A full game fits in a small .pgn file.
 - **Flexible:** Can include one game or many (like a tournament database).
 - **Standard:** Defined by FIDE (chess's governing body) since the 1990s.
- **File Extension:** Saved as .pgn files (e.g., my_game.pgn).
- **How It's Used:** Open in chess software to replay, analyze (e.g., with Stockfish for "eval" scores), or export to images/videos.

PGN ignores things like board visuals (that's for software to render). It focuses on the logic of the game.

3. Overall Structure of a PGN File

A PGN file has two main parts:

- **Headers** (mandatory for structure, but some can be optional): A list of key-value pairs in square brackets [Key "Value"]. These describe the game's metadata (e.g., players, date).
- **Movetext** (mandatory): The sequence of moves in Standard Algebraic Notation (SAN), plus the result. This is the "play-by-play" part.

Optional elements include:

- Comments (in curly braces {comment}).
- Variations (alternative lines in parentheses (move)).
- Annotations (like ! for good move, ? for blunder).

Structure Example (pseudo-code):

```
text
[Header1 "Value1"]
[Header2 "Value2"]
... (more headers)
```

```
1. move1 move1-response 2. move2 move2-response {comment} (variation) ... result
```

- **Multiple Games:** A single .pgn file can have many games, separated by blank lines.
- **Encoding:** Usually UTF-8 for international support.
- **Length:** Short games are ~100 lines; long ones ~500+ with comments.

If a PGN has "eval" values (e.g., {eval: 0.5} from engine analysis), they're treated as comments—software can ignore or use them for graphs.

4. Detailed Explanation: Headers

Headers are at the top, each on its own line in [Tag "Value"] format. There are seven "required" tags (called the "Seven Tag Roster" or STR) for standard compliance, plus optional ones.

- **Required Headers (STR):**
 - **[Event "Event Name"]:** The tournament or match name (e.g., "World Championship 2023").
 - **[Site "Location"]:** Where the game was played (e.g., "New York, USA" or "Online").
 - **[Date "YYYY.MM.DD"]:** Game date. Use "??" for unknown parts (e.g., "2023.?.?.?" if month/day unknown).
 - **[Round "Round Number"]:** Tournament round (e.g., "5" or "-" for casual games).
 - **[White "Player Name"]:** White's player (e.g., "Carlsen, Magnus").
 - **[Black "Player Name"]:** Black's player (e.g., "Nepomniachtchi, Ian").
 - **[Result "Result"]:** Game outcome: "1-0" (White wins), "0-1" (Black wins), "1/2-1/2" (draw), or "*" (ongoing/incomplete).
- **Common Optional Headers:**
 - **[WhiteElo "Rating"] / [BlackElo "Rating"]:** Player ratings (e.g., "2850").
 - **[TimeControl "Time"]:** Time format (e.g., "1800+30" for 30 minutes + 30s increment).
 - **[Termination "Reason"]:** How it ended (e.g., "checkmate", "resignation").
 - **[Annotator "Name"]:** Who added notes.
 - **[Variant "Type"]:** For non-standard chess (e.g., "fischerandom").
- **Explanation:** Headers are like a book's metadata. They don't affect replay but help organize databases. Values are in quotes; unknown is "?".
- **Example:**
text

[Event "Casual Game"]

[Site "Milpitas, California, US"]

[Date "2026.03.05"]

[Round "-"]

[White "TATA"]

[Black "AI (hard)"]

- [Result "1-0"]

5. Detailed Explanation: Movetext (The Moves)

This is the core: a numbered list of moves in SAN, ending with the result.

- **Standard Algebraic Notation (SAN):**
 - **Basic Format:** Piece letter + destination square (e.g., "Nf3" = Knight to f3).
 - **Pieces:** N (Knight), B (Bishop), R (Rook), Q (Queen), K (King). Pawn moves omit letter (e.g., "e4").

- **Captures:** "x" (e.g., "Bxc6").
- **Disambiguation:** If two pieces can go to the same square, add file/rank (e.g., "Nbd2" = Knight from b-file to d2).
- **Castling:** "O-O" (kingside), "O-O-O" (queenside).
- **Promotion:** "=Q" (e.g., "e8=Q").
- **Check/Checkmate:** "+" (check), "#" (mate) — optional in PGN but common.
- **Analogy:** SAN is like shorthand directions: "Knight to f3" instead of full coordinates.
- **Movetext Structure:**
 - Starts with "1." for White's first move.
 - Format: MoveNumber. WhiteMove BlackMove (e.g., "1. e4 e5").
 - Continues sequentially (e.g., "2. Nf3 Nc6").
 - Ends with result (e.g., "1-0").
 - White/Black moves on the same line; if odd moves, Black's is omitted.
- **Comments:** In curly braces {This is a comment.}. Can be anywhere, often for "eval" (e.g., {eval: +0.5} means slight White advantage).
- **Variations:** Alternative lines in parentheses (2. d4 exd4). Nested OK.
- **Annotations (NAGs):** Symbols like "!" (good move), "?" (mistake), "!!" (brilliant), "???" (blunder). Numeric codes (e.g., \$1 = !).
- **Result:** Must match header; placed at end.
- **Exception Handling in PGN:** Software ignores invalid moves but logs errors. Evals are optional comments.

6. Some Examples

Let's use two examples: one simple (your short PGN) and one with comments/variations.

Example 1: Basic PGN (from your first sample, simplified)

This is a quick checkmate game.

text

```
[Event "Chess Game"]
[Site "Local"]
[Date "2026.02.12"]
[Round "-"]
[White "Human"]
[Black "AI (easy)"]
[Result "1-0"]
```

```
1. e4 Nh6 2. d4 Ng8 3. Nf3 Nf6 4. Ng5 Rg8 5. Qf3 Nxe4 6. Qxe4 Rh8 7. Qf4 Rg8 8. Qxf7# 1-0
```

- **Explanation:** Headers describe the game. Movetext shows moves (e.g., "1. e4 Nh6" = White pawn to e4, Black knight to h6). Ends with "#" for checkmate and "1-0" (White wins). No comments here, but your original had evals like {eval: 0.00} which are ignored for replay.

Example 2: PGN with Comments, Variations, and Annotations

A famous opening (Ruy Lopez) with extras.

text

```
[Event "Example Game"]
[Site "Online"]
[Date "2023.10.01"]
[Round "1"]
[White "Player A"]
[Black "Player B"]
[Result "1/2-1/2"]
[WhiteElo "2200"]
[BlackElo "2150"]
```

```
1. e4 {Pawn opening, controls center.} e5 2. Nf3 Nc6 3. Bb5 {Ruy Lopez opening!} a6 4. Ba4 Nf6 5. O-O (5.
d3 {Safer alternative.}) Be7?! {Inaccurate, better is b5.} 6. Re1 b5 7. Bb3 d6 8. c3 O-O 9. h3 Na5 10. Bc2
c5 11. d4 Qc7 1/2-1/2
```

- **Explanation:**
 - Comments: {Pawn opening...} explains moves.
 - Variation: (5. d3 {Safer...}) shows an alternative at move 5.
 - Annotation: ?! means "dubious move."
 - Result: "1/2-1/2" = draw (e.g., agreement).

From my tool call earlier, here's a generated example using Python's chess library:

text

```
[Event "Example Chess Game"]
[Site "Local"]
[Date "2023.10.01"]
[Round "1"]
[White "Player White"]
[Black "Player Black"]
[Result "*"]
```

```
1. e4 e5 2. Nf3 Nc6 3. Bb5 a6 4. Ba4 Nf6 5. O-O *
```

This is an incomplete Ruy Lopez (result "*" means ongoing).

7. How to Read/Write PGN

- **Reading:** Open in text editor or chess app. Moves are sequential; follow numbers.
- **Writing:** Use chess software or Python (e.g., chess.pgn library) to generate.
- **Common Errors:** Mismatched result/header, invalid SAN (e.g., wrong piece). Software validates.
- **Advanced Tips:** For multi-game PGNs, separate with blank lines. Use engines for evals (e.g., Stockfish adds `{[%eval 0.5] [%clk 0:05:00]}` for score/time).

8. Game of the century

[Event "Third Rosenwald Trophy"]

[Site "New York, NY USA"]

[Date "1956.10.17"]

[EventDate "1956.10.07"]

[Round "8"]

[Result "0-1"]

[White "Donald Byrne"]

[Black "Robert James Fischer"]

[ECO "D92"]

[WhiteElo "?"]

[BlackElo "?"]

[PlyCount "82"]

1.Nf3 Nf6 2.c4 g6 3.Nc3 Bg7 4.d4 O-O 5.Bf4 d5 6.Qb3 dxc4 7.Qxc4 c6 8.e4 Nbd7 9.Rd1 Nb6 10.Qc5 Bg4 11.Bg5 Na4 12.Qa3 Nxc3 13.bxc3 Nxe4 14.Bxe7 Qb6 15.Bc4 Nxc3 16.Bc5 Rfe8+ 17.Kf1 Be6 18.Bxb6 Bxc4+ 19.Kg1 Ne2+ 20.Kf1 Nxd4+ 21.Kg1 Ne2+ 22.Kf1 Nc3+ 23.Kg1 axb6 24.Qb4 Ra4 25.Qxb6 Nxd1 26.h3 Rxa2 27.Kh2 Nxf2 28.Re1 Rxe1 29.Qd8+ Bf8 30.Nxe1 Bd5 31.Nf3 Ne4 32.Qb8 b5 33.h4 h5 34.Ne5 Kg7 35.Kg1 Bc5+ 36.Kf1 Ng3+ 37.Ke1 Bb4+ 38.Kd1 Bb3+ 39.Kc1 Ne2+ 40.Kb1 Nc3+ 41.Kc1 Rc2# 0-1

9. Reference

You can refer some of my chess programs on <https://github.com/santakd/chess>